

ASSIGNMENT 1

MMATH FIRST YEAR, 2025

Problems

1. Show that there is an integer multiple of 7 of the form $111 \cdots 1$.
2. Show that $1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n}$ is not an integer for any integer $n > 1$.
3. Show that for all positive real X sufficiently large, there is a perfect square between X and $X + 2\sqrt{X} + 1$.
4. (a) If $2^n - 1$ is a prime, then show that n must be a prime.
(b) A Fermat prime is a prime number of the form $2^{2^n} + 1$. Show that a prime of the form $2^n + 1$ must be a Fermat prime.
5. Show that product of two consecutive positive integers cannot be a perfect square.
6. Suppose α is an irrational number. Show that there infinitely many rational numbers m/n such that

$$\left| \alpha - \frac{m}{n} \right| \leq \frac{1}{n^2}.$$

Hint: Use the pigeonhole principle.

Remark: The above statement is a theorem of Dirichlet and is the first result in the branch of Number Theory called Diophantine Approximation.

7. Determine for what n , $n^4 + 4^n$ is a prime.
8. Let $f \in \mathbb{Z}[x]$ be a non-constant polynomial. Then show that (a) $f(n)$ is composite for infinitely many $n \in \mathbb{N}$;
and (b) the union of the sets of prime factors of $f(n)$ as n varies over $\mathbb{N}$ is infinite.
9. Suppose $(,)$ denotes the *gcd* and $[,]$ denotes the *lcm* of two integers. Show (preferably without using prime factorization) that
 - (i) $a|bc$ iff $\frac{a}{(a,b)}|c$,
 - (ii) $(a, [b, c]) = [(a, b), (a, c)]$,
 - (iii) $[a, (b, c)] = ([a, b], [a, c])$
 - (iv) $(a, b) = (a + b, [a, b])$.
10. If $ad - bc = 1$, show that $(a + b, c + d) = 1$.
11. Show that $\binom{2n}{n}$ is divisible by $n + 1$.
12. If $2^n + 1$ is a prime, show that n itself a power of 2.
13. A *perfect number* is a positive integer which equals the sum of all its divisor except itself; e.g., $6 = 1 + 2 + 3$. Prove that an even number n is perfect iff n is of the form $n = 2^{p-1}(2^p - 1)$, where p as well as $2^p - 1$ are primes.

Remark: This is a theorem of Euclid and gives a complete and explicit description of all even perfect numbers. Apparently, the oldest unsolved problem in Mathematics is to determine if there exists an odd perfect number. See the discussion here: <https://mathoverflow.net/questions/27075/what-is-the-oldest-open-problem-in-mathematics>.